	ORM	PRM	Majority Voting
% Solved (Best-of-1860)	72.4	**78.2**	69.6

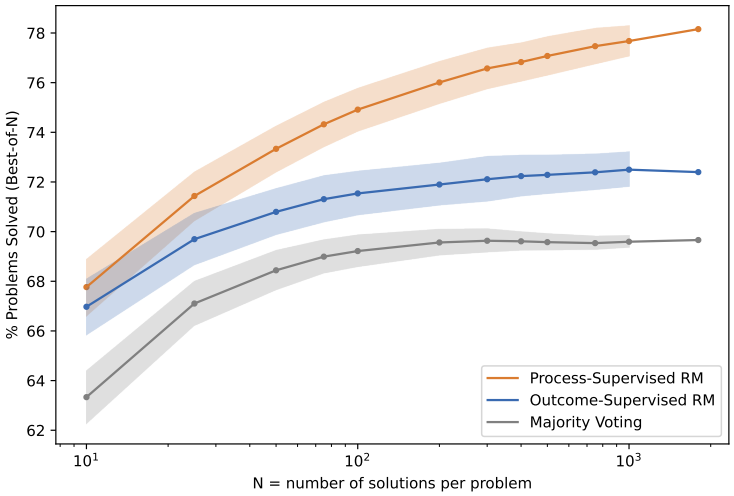
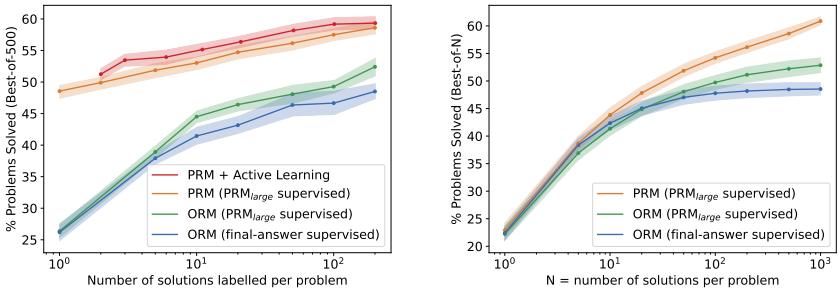


Figure 3: A comparison of outcome-supervised and process-supervised reward models, evaluated by their ability to search over many test solutions. Majority voting is shown as a strong baseline. For $N \leq 1000$, we visualize the variance across many subsamples of the 1860 solutions we generated in total per problem.

3 Large-scale Supervision

We train the large-scale PRM using the step-level labels in PRM800K. To ensure the large-scale ORM baseline is as strong as possible, we train on 100 uniform samples per problem from the generator. This means the ORM training set has no overlap with PRM800K, and it is an order of magnitude larger. Although these two training sets are not directly comparable, each represents our best attempt to advance the state-of-the-art with each form of supervision. We note that training the ORM solely on PRM800K solutions would be problematic, since our active learning strategy has heavily biased the dataset towards wrong-answer solutions. We did explore training the ORM on a superset of PRM800K solutions, by mixing in uniformly sampled solutions, but we found that this did not improve ORM performance.

Figure 3 shows how the best-of-N performance of each reward model varies as a function of N. Since majority voting is known to be a strong baseline (Wang et al., 2022; Lewkowycz et al., 2022), we also include this method as a point of comparison. While the ORM performs slightly better than the majority voting baseline, the PRM strongly outperforms both. Not only does the PRM reach higher performance for all values of N, but the performance gap widens as N increases. This indicates that the PRM is more effective than both the ORM and majority voting at searching over a large number of model-generated solutions.



(a) Four series of reward models trained using different data collection strategies, compared across training sets of varying sizes.

(b) Three reward models trained on 200 samples/problem using different forms of supervision, compared across many test-time compute budgets.

Figure 4: A comparison of different forms of outcome and process supervision. Mean and standard deviation is shown across three seeds.

We experimented with using RM-weighted voting (Li et al., 2022; Uesato et al., 2022) to combine the benefits of the PRM and majority voting, but this did not noticeably improve performance. We use a specific subset of the MATH test set for evaluation, which we describe in Appendix C. We further break down these results by problem difficulty in Appendix G.

4 Small-scale Synthetic Supervision

We find that the PRM outperforms the ORM at large-scale, but this result alone paints an incomplete picture. To better compare outcome and process supervision, there are two confounding factors that must be isolated. First, the training sets for the ORM and the PRM are not directly comparable: the PRM training set was constructed using active learning, is biased towards answer-incorrect solutions, and is an order of magnitude smaller. Second, the final-answer grading will provide positive labels to spurious solutions that reach the correct final answer despite incorrect reasoning. This could damage ORM performance, an effect we may or may not want to attribute to outcome supervision more generally.

Due to the high cost of collecting human feedback, we cannot easily ablate these factors using human labelers. We instead perform the relevant ablations by using the large-scale PRM to supervise smaller models. This setup enables us to simulate a large amount of data collection at a modest cost. For the remainder of this section, we refer to the large-scale PRM from Section 3 as $\text{PRM}_{\text{large}}$.

4.1 Process vs Outcome Supervision

We now conduct a direct comparison of outcome and process supervision. We first sample between 1 and 200 solutions per problem from a small-scale generator. For each dataset, we provide three forms of supervision: process supervision from $\text{PRM}_{\text{large}}$, outcome supervision from $\text{PRM}_{\text{large}}$, and outcome supervision from final-answer checking. The choice of supervision is the only difference between these three series of reward models, which are otherwise trained on identical datasets. See Appendix H for more details about how $\text{PRM}_{\text{large}}$ is used for outcome and process supervision.

In Figure 4a, we evaluate each reward model by its best-of-500 selection. We see that process supervision significantly outperforms both forms of outcome supervision at all data collection scales. In Figure 4b, we evaluate the best reward model from each series by its best-of-N performance across different values of N. We see that using $\text{PRM}_{\text{large}}$ for outcome supervision is noticeably more effective than final-answer checking. This can be explained by the fact that $\text{PRM}_{\text{large}}$ provides better supervision for solutions that reach the correct final answer using incorrect reasoning.

It is not clear whether supervision by $\text{PRM}_{\text{large}}$ or by final-answer checking represents the more appropriate outcome supervision baseline. While final-answer supervision is more explicitly outcome based, its main weakness — the existence of false positives — is arguably over-emphasized in the MATH dataset. Outcome supervision by $\text{PRM}_{\text{large}}$ better represents outcome supervision in domains that are less susceptible to false positives. We consider outcome supervision by $\text{PRM}_{\text{large}}$ to be the more relevant baseline, but we encourage the reader to draw their own conclusions.

4.2 Active Learning

Finally, we investigate the impact of active learning. We train a small-scale reward model, $\text{PRM}_{\text{selector}}$, on a single sample from each problem, and we use this model to score 1000 samples per problem. To train each of our larger reward models, we select N samples per problem such that 80% are the most convincing (according to $\text{PRM}_{\text{selector}}$) wrong-answer samples, and 20% are the most convincing samples that remain (right- or wrong-answer). We score the selected samples with $\text{PRM}_{\text{large}}$ and train on those scores. This process ensures that all samples are relatively convincing under $\text{PRM}_{\text{selector}}$, that a large fraction are known to contain at least one mistake, and that our overall dataset is not too heavily biased toward wrong-answer solutions. Performance of this data labelling scheme is shown in Figure 4a. By comparing the slopes of the line of best fit with and without active learning, we estimate that this form of active learning is approximately 2.6x more data efficient than uniform data labelling. We note that the model trained on the largest active learning dataset (200 samples per problem) appears to slightly underperform the expected trend line. Our best explanation for this observation is that 200 samples represents a significant fraction of the overall selection pool (1000 samples) and that this